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Bachelor of Computer Applications

Semester – V

Business Analytics

Assignment : 2

Title: Superstore Sales Performance Analysis Using Power BI

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Superstore Sales Performance Analysis Using Power BI

Problem Statement

As a Data Analyst working with a retail superstore chain, the task is to evaluate overall sales performance across product categories, sub-categories, regions, individual products, and time. Management needs a single, interactive dashboard that summarizes key performance indicators — total sales, total profit, quantity sold, and profit margin — and allows them to drill into performance by category, region, sub-category, product, and year, in order to identify top-performing segments and support data-driven business decisions.

Methodology / Steps Followed

Step 1: Data Collection

- Obtained the Superstore sales dataset containing transaction-level records.
- Identified key fields: Order ID, Order Date, Region, Category, Sub-Category, Product Name, Sales, Profit, and Quantity.

Step 2: Data Import

- Opened Power BI Desktop and imported the Superstore dataset using Get Data.
- Loaded the data into the Power BI data model for report building.

Step 3: Data Cleaning and Transformation

- Checked the dataset for missing values and inconsistent entries.
- Verified data types for Order Date, Sales, Profit, and Quantity fields.
- Created a calculated measure for Profit Margin ($\text{Total Profit} \div \text{Total Sales}$).
- Ensured Category, Sub-Category, Region, and Product Name were recognized as text/categorical fields for grouping.

Step 4: Power BI Interface Exploration

- Used the Data pane to select the required fields for each visual.
- Used the Visualizations pane to build KPI cards, column charts, bar charts, a line chart, and a stacked column chart.
- Formatted axis titles, legends, and data labels for readability.
- Arranged all visuals on a single report page to form a cohesive dashboard.

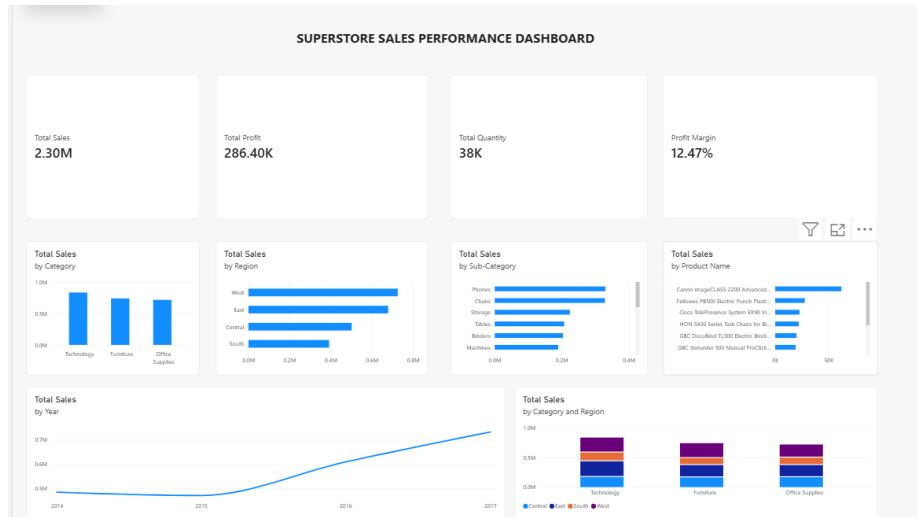
Step 5: Dashboard Development

The following Superstore Sales Performance Dashboard was created:

- Total Sales – KPI card showing overall sales revenue.
- Total Profit – KPI card showing overall profit earned.
- Total Quantity – KPI card showing total units sold.
- Profit Margin – KPI card showing overall profit as a percentage of sales.
- Total Sales by Category – Column chart comparing Technology, Furniture, and Office Supplies.
- Total Sales by Region – Bar chart comparing West, East, Central, and South regions.
- Total Sales by Sub-Category – Bar chart comparing Phones, Chairs, Storage, Tables, Binders, and Machines.

- Total Sales by Product Name – Bar chart ranking individual top-selling products.
- Total Sales by Year – Line chart showing the sales trend from 2014 to 2017.
- Total Sales by Category and Region – Stacked column chart breaking down each category's sales by region.

Output



Superstore Sales Performance Dashboard — Power BI Desktop

Explanation of the Output

The dashboard gives a complete, at-a-glance view of the superstore's sales performance, combining summary KPIs with category-, region-, product-, and time-based breakdowns.

- **Total Sales:** The KPI card shows overall sales of 2.30M, representing the total revenue generated across all orders.
- **Total Profit:** The KPI card shows a total profit of 286.40K earned across all transactions.
- **Total Quantity:** The KPI card shows 38K units sold in total.
- **Profit Margin:** The KPI card shows an overall profit margin of 12.47%, calculated as total profit divided by total sales.
- **Total Sales by Category:** The column chart shows Technology generating the highest sales, followed closely by Furniture and Office Supplies.
- **Total Sales by Region:** The bar chart shows the West region leading in sales, followed by East, Central, and South.
- **Total Sales by Sub-Category:** The bar chart shows Phones as the top-selling sub-category, followed by Chairs, Storage, Tables, Binders, and Machines.
- **Total Sales by Product Name:** The bar chart ranks individual products by sales, highlighting items such as the Canon imageCLASS 2200 Advanced copier and the Fellowes PB500 Electric Punch Plastic Comb Binding Machine among the top sellers.
- **Total Sales by Year:** The line chart shows sales were relatively flat from 2014 to 2015 and then rose steadily through 2016 into 2017, indicating consistent year-over-year growth.
- **Total Sales by Category and Region:** The stacked column chart breaks down each category's sales by region (Central, East, South, West), showing that West and East contribute the largest shares within every category.

Conclusion

The Superstore sales dataset was successfully imported into Power BI and transformed into an interactive performance dashboard. The dashboard consolidates key metrics — total sales, total profit, quantity sold, and profit margin — alongside category, region, sub-category, product, and year-wise breakdowns. The analysis shows that Technology is the leading sales category, the West region is the strongest performer, Phones is the top sub-category, and sales have grown steadily since 2015. This dashboard equips management with a clear, data-driven view for monitoring performance and identifying growth opportunities.

Learning Outcomes

Through this experiment, I learned:

- How to import and prepare a retail sales dataset in Power BI.
- How to build KPI cards to summarize key business metrics such as sales, profit, quantity, and margin.
- How to create calculated measures such as profit margin.
- How to use column, bar, line, and stacked column charts to compare performance across categories, regions, products, and time.
- How to design a single-page, multi-visual dashboard that communicates business insights clearly.
- How to interpret sales and profitability trends to support management decision-making.

Github link : <https://github.com/varshithramesh/Business-Analytics-Projects>